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Dear Editor,

We submit our manuscript entitled “*Adaptive Multi-Representation Fusion via Dual-Channel PCNN with Multi-Scale Convolution for Vibration Signal Fault Diagnosis*” for consideration in *Digital Signal Processing*.

Single-representation methods for vibration-based fault diagnosis capture only partial signal characteristics, and existing fusion strategies lack adaptivity in regulating heterogeneous representations. We propose a two-stage framework: an adaptive weighted dual-channel pulse-coupled neural network (AW-DPCNN) that fuses STFT spectrograms and GADF images via contrast-guided dynamic weighting, paired with a multi-scale convolution and channel attention enhanced VGG16 (MSCA-VGG16) classifier. The main contributions are:

1. **AW-DPCNN fusion mechanism:** An adaptive fusion mechanism is proposed that integrates heterogeneous STFT spectrograms and GADF images via contrast-guided dynamic weighting and dual-channel convolutional stimuli, adaptively regulating per-channel contributions to suppress destructive interference and enhance feature complementarity.
2. **MSCA-VGG16 classification network:** A dedicated classifier is designed with multi-scale convolution for capturing fault patterns at multiple temporal and spectral granularities, squeeze-and-excitation channel attention for dynamic frequency-band recalibration, and a compact embedding head for enhanced intra-class compactness and inter-class separability.
3. **Comprehensive experimental validation:** Extensive experiments on the CWRU bearing dataset, including backbone and representation comparisons, systematic ablation studies, and noise robustness evaluations, with cross-sensor generalization to fan-end data (97.45%) and cross-sampling-rate robustness at 48 kHz (94.81%).

This work aligns closely with the scope of *Digital Signal Processing*. The AW-DPCNN mechanism advances time-frequency representation fusion through adaptive, contrast-driven weighting, contributing to signal processing theory with relevance to multi-sensor systems. The MSCA-VGG16 classifier extends spectral analysis via multi-scale feature extraction with channel-wise attention. Beyond bearing diagnostics, the framework generalizes to heterogeneous signal representation fusion in acoustic monitoring, speech processing, and industrial condition monitoring, underscoring its broader impact on data-driven signal processing.

This manuscript is original, has not been published elsewhere, and is not under consideration by another journal. All authors have approved the submission. We declare no competing interests.

Thank you for your time and consideration.

Sincerely,

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